Term Project
Business and Logistics plan for “Fresh Veggie” (FV)

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You and your team will develop a business and logistics plan for a new company: "*The Fresh Veggie (or FV)*". FV has identified a business opportunity in customer-direct sales of fresh vegetables and fruits to customers in the **Metropolitan Region of Santiago, Chile**. Santiago has one of the highest per capita income in Chile, concentrating also a significant percentage of the population in the country.

Under the envisioned market strategy, the customer will place orders based on an app or accessing to the website of the company. Orders will be received in the service center from 8:00 AM to 9:00 PM, every day of the week. Once the order is received and processed, the order information is routed to the distribution team, which oversees putting the order together at a distribution center (DC) and making the delivery no later than **20 hours** after the order is received (see Appendix A for an example of the dispatching of the orders received).

FV will make bulk purchases of products either from packing facilities located at the production sites (both local and imported) or locally, from the wholesale produce market. The prices of the products vary according from where they are acquired. For instance, the prices at the packing facilities (at the source) are usually lower than those found at the terminal markets; however, the transportation costs for the products bought at the source will be significantly higher than those for products purchased at the local wholesale market. FV will be responsible for transporting the products purchased from the supply point (either packing facilities or local wholesale market) to the local Distribution Center (DC), from where the products are packaged into individual orders and distributed directly to the final customer, and/or by dark stores located within the service region and/or by a mobile distribution center that consists on trucks that need to be parked (in paid parking areas, so this will be part of the cost to consider) and replenish small vehicles for the delivery to the households (e.g. bikes, motorcycles, robots, etc.). You will have to evaluate the different alternatives and select the most convenient.

The profits that the company make are directly influenced by the price paid for the product, the logistics costs, the service level offered, and the associated food waste. Thus, your task is to prepare a comprehensive plan for the design of the logistics network that will enable the direct sales and delivery of the targeted products to the consumer in the most efficient manner. In preparing your plan, you must consider that the delivery time is influenced by the logistics infrastructure and distribution strategies employed by the company - such as the location of the DC(s), dark stores, mobile distribution centers, inventory held and characteristics of the transportation fleet.

Products to be sold, potential sources and prices

Several targeted products and producing regions have been initially identified to start the operations of FV. Each of producing regions can grow one or more of the targeted products under specific growing/harvesting windows. These windows are determined by the local climate and geographic location of the producing region. It is assumed that demand is constant for each of the targeted products throughout the year. However, a single region will not be able to provide a year-round supply of any product.

Demand for each of the targeted products is determined by the average daily consumption of each product in CHILE. However, despite the constant daily demand, it is assumed each household will place orders for one or more products according to a policy that will minimize its total costs. Students need to find information related to the consumption of the selected products.

To simplify the problem, FV will target the Metropolitan area of Santiago to distribute a basket of **FIVE** fresh products: **Roma Tomatoes, Cauliflower, Broccoli, Asparagus, Green Bell Pepper.**

The potential sources of supply include the following countries, but this depends on the past imports reported by Chilean Customs from abroad, and for the domestic case based on the real places where are produced:

- In the US: San Joaquin, Merced, and Fresno counties in California (Calif); Miami-Dade, Palm Beach and Hillsborough Counties in Florida (FL)
- In Mexico: Sinaloa (Sin), Chihuahua (Chih) and Bajio region (Baj) (which includes the states of Michoacan, Guanajuato, Queretaro)
- In Peru: Callao (Call), Piura (Piur), Illo (Illo), Arequipa (Areq)
- In Ecuador: Guayas (Guay), El Oro (Oro), Los Ríos (Rios)
- In Bolivia: La Libertad (Lib), Cochabamba (Coch),
- In Argentina: Salta (Sal) and Jujuy (Juj), Mendoza (Mdz)
- In Spain: Granada (Gran), Córdoba (Cor), Sevilla (Sev), Murcia (Mur), Andalucía (And)
- In Chile: Santiago (Metropolitan Region of Santiago (Scl), Valparaiso Region (Vap), Maule Region (Maul), Coquimbo Region (Coq).

Moreover, products can be also acquired from wholesale markets in Santiago such as La Vega and Lo Valledor. The information for prices for imports of the different products and countries in 2025 are given from Chilean Customs which has been processed and complemented by the company Datasur. According to past imports, you can determine also which are the actual sources of supply. The dataset does not specify the variety of the product, but you can search directly on the website of Customs or other sources to investigate and make some assumptions to define the price of your product. Furthermore, you may assume, in the case of imports, that the price to acquire these products from the supplier is the import price (given on a CIF basis). For the case of domestic products, the price will be given at the shipping point, or the terminal market (as part of the analysis, you need to decide which is a better option). You can also assume that the production capacity of the suppliers is unlimited within their growing window. If you don´t have the specific region within a country from which products are sourced, you can estimate it by providing an assumption, considering the traveling times accordingly. Finally, you can assume that the wholesale markets offer the product year-round without growing window limitations.

You also can assume that the price paid by the final consumer is the average price reported by Chilean supermarkets such as Lider or Jumbo. For this, you need to check the prices on the website and observe or predict the behavior of these prices throughout the year. Use the prices for bulk, non-organic, and open-field products. For consistency, we will calibrate the prices among the different teams before the final report. To standardize demand estimates, make the following assumptions:

- For tomatoes, assume that the final price corresponds to Roma Tomatoes, and this is the 50% consumption of all the varieties if you don't find the precise information.
- For green bell pepper, assume that 70% of the consumption of all the varieties of bell pepper.
- The consumption for the rest of the products corresponds to 100% of the reported per-capita consumption (without considering a specific variety).

It is assumed that the customers require the constant availability of the products throughout the year, and as part of your task, you must design a purchasing plan for each product for the entire year.

To have the same basis of comparison assume that the population in 2025 is the same as the population reported in the 2024 census. By using the proper sources and methodology, you also have to estimate the average number of people per household.

Transportation Issues

Regularly, when FV places an order to a foreign supplier, the supplier is responsible for loading and transporting the container to the border the same day the order is placed. You can assume that any product requires 1 day of packing after harvest, but you need to estimate the time required to transport the product to the border facility, and then from the border to the DC. Products are transported by truck or vessel according to the location of the supplier. If you source products directly from a local field, you have to also consider the same time for packing after harvest (1 day) and the required traveling time to your DC. From foreign suppliers you can either use road/rail transport mode (rail if the service is available), maritime transport or air transport. For air transport, all the products are received in the Arturo Merino Benitez Airport in Santiago. For maritime transport, assume that the products are sent to either the port of San Antonio or the port of Valparaíso.

Notice that transit times are stochastic. For this reason, you will have to estimate the average transit time of each leg of the trip and the associated variability (as a rule of thumb, you can estimate the standard deviation as a 5% of the average transit time for trucks or rail and 10% for maritime vessels, and 1% for an airplanes). Please note that as a part of the transit time, average driving speed by truck, for long-haul distances, may differ from urban (local) distribution of the products.

Additional Market Issues

FV will deliver fresh produce directly to the household within the Metropolitan Region of Santiago area for a flat delivery fee **of $2,5 per order**. You can assume that the upper limit market potential corresponds to a **20%** of the overall estimated consumption in the targeted area (Metropolitan Region of Santiago). This is the upper limit of the potential market and it depends on the logistics network designed by each team if you can satisfy it. You can further assume that the household consumption of each product is constant throughout the year. For instance, if the annual demand for product A is 450 kgs. per household, then the daily demand would be 450/365 kgs. per household.

You can also assume that each one of the households will have an ordering policy in place that is independent of the rest of the households. For instance, a household may have a policy to place an order every week to replenish the inventory of all the products, while another household may place an order every other day. You can use a Poisson distribution to model the placement of orders to FV. The determination of the parameters (λ) of the distribution is part of the analysis that you need to perform.

Another constraint is that the service region has to be delimited by **15 kilometers from a DC and 5 kilometers from a dark store, and there is no limit if a mobile distribution center is used as it can be allocated to specific parking areas where it replenishes the fleet of vehicles for last mile delivery. There is no restriction to serve the dark stores from a DC neither to serve mobile distribution centers.** Hence, the actual market captured is highly dependent on the underlying distribution practices. Furthermore, you can assume that you need to provide at least 95% fill rate to your customers, but the exact fill rate will be determined based on a profit maximization study you will perform for FV.

Perishability, Consumer order size

All the targeted products are perishable and need to be maintained in the "cold chain" to preserve their shelf lives. Even if maintained in optimal cold storage conditions, for every day that elapses after harvest, the shelf life of these products decreases. In general, it is assumed that the remaining shelf life for each product is given by the following functions:

$$Decay_t = \frac{k_2}{T}\left(1 - e^{-\frac{k_2}{T}t}\right)$$

$$\%\ Remaining\ Shelf\ life = T\left(1 - \sum_{k=0}^{T} Decay_t\right)$$

Where the parameters k_2 and T depend on each product and t is the number of days after harvest. Larger values of T imply higher total shelf life. The corresponding values for T and k_2 for the products to target are given in the table below.

Table 1. Shelf Life of the produces

Product	k2	T
Roma Tomatoes	2	14
Cauliflower	2	12
Brocolli	2	12
Asparagus	2	15
Green B, Pepper	2	16

An example of this function is given in Figure 1.

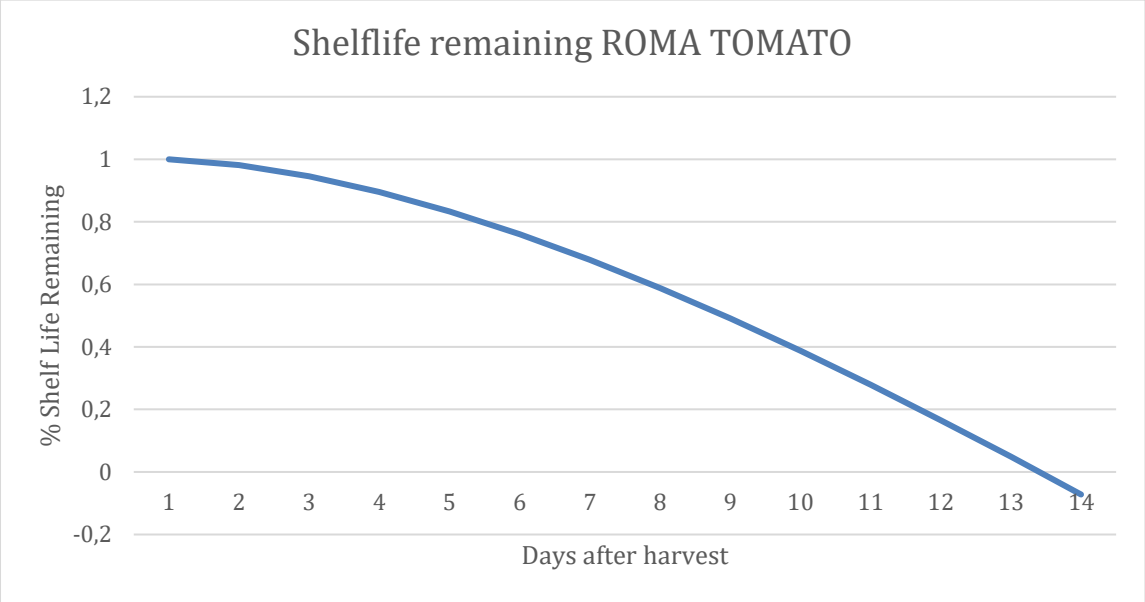

Figure 1. Roma tomato remaining shelf life as a function of the days after harvest.

To determine the inventory policy, you need to consider the perishability of the products. To do this, you need to take the lead time required for the products, the order frequency and desired inventory levels to design an ordering policy that reduces product waste and total cost. If waste is generated, assume that its cost corresponds to the total cost incurred to get the product to where the waste occurs. **Inventory holding costs corresponds to a 15% rate per year.**

As part of the project, you need to determine the order size and the frequency of orders of the customers. Since each household will minimize total ordering cost, it is assumed that they incur in addition to the fixed ordering cost of **$2,5** and a **15% rate** associated with the inventory holding cost per year. In determining the order placed by a customer, you also must consider the perishability of the products, keeping in mind that if a product is not consumed within the corresponding shelf life it will be wasted.

For quality reasons, and as a policy of the company, to guarantee freshness, no product can be delivered to the customer with less than **35% of remaining shelf life**. Those products that do not fulfill this requirement are considered waste, and need to be properly disposed of.

Finally, for the last mile strategy you need to determine the type of vehicles and fleet size to be used. You will assume that you will hire the transport services to external transport providers, either in the spot market or under a dedicated fleet. Or alternatively, you can acquire your own fleet and assume that

the cost will be paid over a amortization cost of 5 years. You can also lease or rent the fleet of vehicles instead of acquiring it. You have to evaluate which strategy is more convenient.

Approach and Deliverables

Your task is to act as an external consulting company that is proposing to develop a comprehensive market/logistics analysis for FV. This includes a recommendation for the market to capture, developing the purchasing and ordering strategy and the associated logistics to maximize profits and minimize waste. This also includes the determination of the products' sources and associated harvesting timing of each product, the location and the size of the DCs and dark stores according to your network design strategy, the layout design of the facilities (at least in terms of dimension and main areas), long-distance and local (last-mile) transportation strategies, composition of the distribution fleet and the ordering policies.

The project consists of three reports to be prepared and delivered during the semester, where the last one is the final consolidated report containing all information of the analyses performed. Some of your tasks will include, but are not limited to:

- Developing a sound strategy to convince FV to retain you as an external consultant. This includes developing the objectives of the business plan as well as the methodology to obtain a competitive and environmental market/logistics strategy. Thus, part of the analysis will explore those issues related to the impact on the environment that the resulting strategies will have.
- Searching for relevant information on the product production, transportation rates and times, as well as any other necessary data.
- Developing adequate models to solve the problem.
- Writing concise reports that contain a description of your approach to the solution of the problem, problem assumptions, data gathered, models and heuristics used, along with your conclusions and recommendations. You also need to include an excel file that summarizes your computations in an organized way.

You will be expected to perform an analysis to identify the source regions for each of the products. Make sure that you consider the perishable nature of the products being distributed into the analysis. Please have in mind that while the main goal is to maximize profits, you will also need to consider food waste and environmental effects of the strategy followed.

Basically, the outcome of the final project must answer the following questions:

1. What is the market that you will serve?
2. Where and how to source the products?
3. What will be the size of your orders and the frequency?
4. What will be your supply chain network strategy? LDC that directly serve the households? Or a RDC supported by Dark Stores?
5. According to your supply chain network strategy, how many DC locations and where to locate them?
6. How should the products reach the customer?
7. How should you develop a flexible last-mile strategy that considers the variability of the demand?
8. Which is the best strategy to maximize profits based on a total landed cost approach?

Some of the factors to be reported in the final project include:

- Identification of customer base requirements (Demand assumptions)
- Facility location problems
- Main transportation routes (including transportation distance, average time and variability)

- Inventory policies
- Alternatives comparison
- Disposal of perished product
- Last-mile delivery details
- Carbon footprint
- Profit maximization techniques

You should bear in mind that the costs associated with providing the services include:

- Cost to establish and operate a Distribution Center (DC); use the appropriate amortization techniques (if needed)
- Cost of inventory (in field/cycle inventory cost per DC, in-transit inventory cost and safety stock cost)
- Cost of transportation (long-haul traveling and last-mile distribution)

All deliverables will consist of a written report (PDF) in spanish accompanied by an Excel file that supports your results and supporting computations. A standard format for each deliverable will be provided by the instructor for you to fill in with the data from your solution. For each deliverable, you are required to prepare a 20-minute PowerPoint presentation in Spanish, delivered according to the program of the course

Please keep in mind that every time you deliver a new report, you can change your previous decisions based on the new information available. Therefore, it is very important that you pay attention to the solutions presented by the rest of the teams and compare and contrast their solutions to yours. The final grade is based on part of the ranking of the proposed solutions from the different teams, so your team should pay attention and scrutinize the results of the rest of the teams to make sure other teams are presenting valid ideas and solutions.

The content of the four reports should be as follows:

1. **FIRST DELIVERABLE. Overall strategy analysis and preliminary logistics network design:** In this report/presentation **you will take the role of a consultant that is presenting a proposal to a potential customer for the development of the overall project**. Thus, you need to prepare a professional proposal from your consulting team to FV. The main objective is to convince FV that you know how to address the problem, that you have the appropriate team to do this and that your price is competitive. Thus, the consulting team needs to define the objectives, the scope of the project, estimate market potential and potential product sources (suppliers) for each product. It is necessary to gather real information related to the harvesting times for each product and each source, as well as the corresponding costs for the purchasing and transportation (long-haul trip) of the products, preliminarily selecting the sources/suppliers, and the related information that may allow your time to have an estimate of your potential earnings. As part of this report, you will obtain the seasonality and costs for each one of the products and regions considered and use that information to have an estimate of your potential earnings. In terms of the market to service, even though it is known that the Metropolitan Region of Santiago is the target market, it will also be necessary to define the specific areas that will be serviced to maximize profits. You have to propose a preliminary network design of the company, indicating the potential location of your DCs and Dark Stores (the latter if you decide to use this strategy).

2. **SECOND DELIVERABLE. Logistics Network Design and Inventory Policy.** You and your team will obtain all data necessary to provide a more detailed network design either including only DCs or a combination of dark stores, as well as the layout design of the facilities indicating the required area which has to be consistent with your estimated costs. Moreover, you have to provide and inventory policy to satisfy the assumed demand restricted on the delivery times. In other words, the team will consider the potential perishability curve, which causes the shelf life of the product to decrease over time. Then, the inventory policy should be aligned with this curve. You must decide whether to use a consolidated policy (when you supply several products from the same source), or an individual policy for each product, considering the harvesting seasonality according to the origin of the products. Finally, you will prepare a preliminary transportation strategy to be used. Notice that we will assume that there are no constraints with respect to mixing different products in a container (in practice you need to consider the compatibility of products to determine if they can be shipped in the same container).

3. **THIRD DELIVERABLE. Last mile Distribution Strategy and Final Report.** In this phase, in addition to refining the long-haul transportation and sourcing of the products to the DCs in your network, you will design the last mile strategy in which you have to determine the detailed transportation strategy for transporting the product from the DCs and a preliminary strategy to transport the product from the DC to the customers (last mile). You have to indicate the type of vehicles and transportation strategy (own fleet or external) to be used. All previous phases should be reconsidered, all assumptions should be stated, and solid arguments should be presented for the decisions that you and your team make. You can refine previous decisions and prepare a final report and presentation that you will provide to your customer. You should include a strategy for the maximization of profits. Thus, some of the points to consider are:

   • Exact definition of boundaries of regions to be serviced

   • Alignment of supply chain policies with market strategy

   • Location (and type) of distribution centers and dark stores and their layout.

   • Composition of distribution fleet and routing.

   • Inventory/ordering policies.

Except for the final report, each one of the reports is limited to 10 pages in addition to appendixes. The final report is limited to 20 pages in addition to appendixes. You will have to provide an excel file (the template will be provided) with the computations required for each deliverable. Notice that every deliverable should summarize previous decisions - highlighting the most important aspects. Not including all the relevant information in a report is penalized.

The best project (only one team per university) will receive a grade of 7.0 and the grades will decrease according to a ranking determined by the professors of the course.

Suggested outline for your final report is:
1. Executive summary: a stand-alone section that provides a clear statement of the situation, a problem description, the goals for the project, a summary of the analysis, results, and general recommendations (Gel, 2007).

2. Introduction: its objective is to provide readers with more details about the objectives and the background of the project. Some elements to consider when writing an introduction are the following (Michaelson, 1990):

a. The purpose of the report
b. The scope of the manuscript
c. The motivation to the problem
d. The rationale of the project

3. Problem description and method of analysis: it embodies any assumptions you make in the model parameters, the information you've gathered, etc. It also includes a detailed description of the problem, an explanation of the methodology followed, etc.

4. Results: All results that you consider relevant and that you will refer to in the concluding section must be included here. Revise for complete and accurate computations; make use of tables and graphs for presenting results.

5. Recommendations and Conclusions: A conclusion not only reviews the results, as a summary section would do, but it also interprets them. A conclusion section should point out what is important from the results, what is significant, and why the results are valid. It might also include recommendations to the management of a company, and how these recommendations are tied to the results previously obtained. A recommendation section is your opportunity to show your expertise in the topic, take advantage of the opportunity.

Grading:

0,30*(Deliverable 1)+ 0,30*(Deliverable 2) + 0,40*(Deliverable 3)

Deliverables_grade: 0,70*(Report) + 0,25*(Dissertation) + 0,05*(Pair groups evaluation)

Sources of information

-Dataset from Chilean Customs related to imports of products.
(given to the students)

-ODEPA Agricultural Policies and Studies Agency in Chile: you have to search for information related to consumption of the products
https://www.odepa.gob.cl/

-INE National Statistics Institute in Chile: for information related to population.
https://www.ine.gob.cl/

-Terminal Market: La Vega and Lo Valledor in Santiago
https://mercadolovalledor.cl/

-Supermarkets: Líder (Walmart) and Jumbo
https://www.lider.cl/supermercado/
https://www.jumbo.cl/

APPENDIX A

Limitation on orders fulfillment

The operating hours for FV (including the distribution center and delivery department), will follow continuous shifts of 8 hours. All the orders received between 8:00 am and prior to 9:00 pm, will be delivered no later than 20 hours after receiving the call. Once the order is received, it takes two hours for the distribution center to prepare it for delivery. The call center is outsourced and not be part of the analysis. The exact time to deliver the order will depend on the distribution strategy and the corresponding transit times.

The following table gives the outline of a delivery strategy that uses only one shift for the DC and an internal delivery fleet that considers the 20 hours filling contrasting of orders.

Table 2 Orders fulfillment illustration

Order Received (using 24 hours)	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Earliest shipping time (today)	10	11	12	13	14									
Earliest shipping time (tomorrow)					8	8	8	8	8	8	8	8	8	8
Latest shipping time (today)	14	14	14	14	14									
Latest shipping time (tomorrow)					8	9	10	11	12	13	14	15	15	15